Requirement Analysis – EV Charge and Range Analysis Dashboard

1. Dataset Description

The dataset used in this project contains information about Electric Vehicles such as vehicle brand, model, battery capacity, range, price, top speed, charging type, and charging station locations. The dataset was collected from publicly available EV data sources and stored in CSV format before further processing.

2. Data Preprocessing Steps

- Collected EV datasets containing vehicle specifications and charging station information.
- Imported the dataset into a SQL database for structured storage.
- Checked for missing or null values in important columns such as range, price, and battery capacity.
- Removed duplicate records to ensure data consistency.
- Standardized column names and data formats for easier analysis.
- Converted categorical fields such as brand, body style, and powertrain into consistent values.
- Created calculated fields such as efficiency metrics and price comparisons.
- Filtered and structured data to prepare it for Tableau visualization.

3. Tableau Dashboard

The Tableau dashboard provides interactive visualizations for EV analysis including:

- EV Charging Stations by Region
- EV Charging Stations Map of India
- EV Brand and Model Comparison
- Top Speed Comparison of EV Brands
- Price Comparison of EV Models
- Top Efficient EV Brands

Inserted Dashboard Screenshot Below:

4. Tableau Story

The Tableau Story presents a sequence of visual insights explaining EV data trends and comparisons. It highlights key findings such as charging infrastructure distribution, vehicle performance comparison, and price-to-efficiency insights.

Insert Tableau Story Screenshot Below:

Story Of Electric Cars In India

Chargin Station India | Charging Station By Region And | Price Of Electric Cars By Different | Different Brands And Number Of

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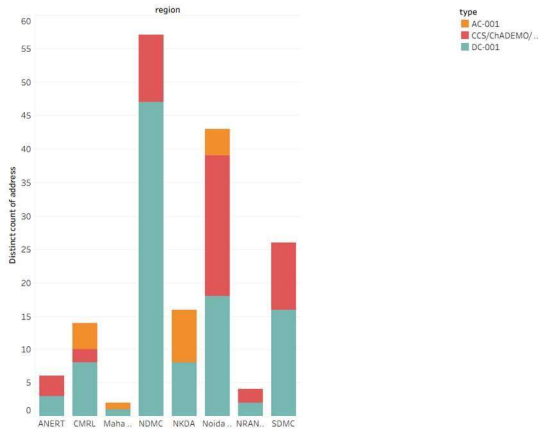
Chargin Station India | Charging Station By Region And | Price Of Electric Cars By Different | Different Brands And Number Of



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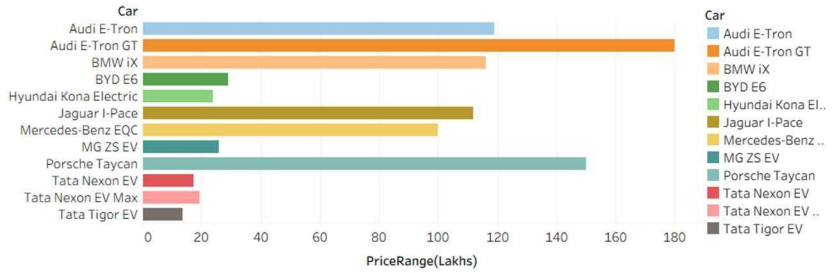

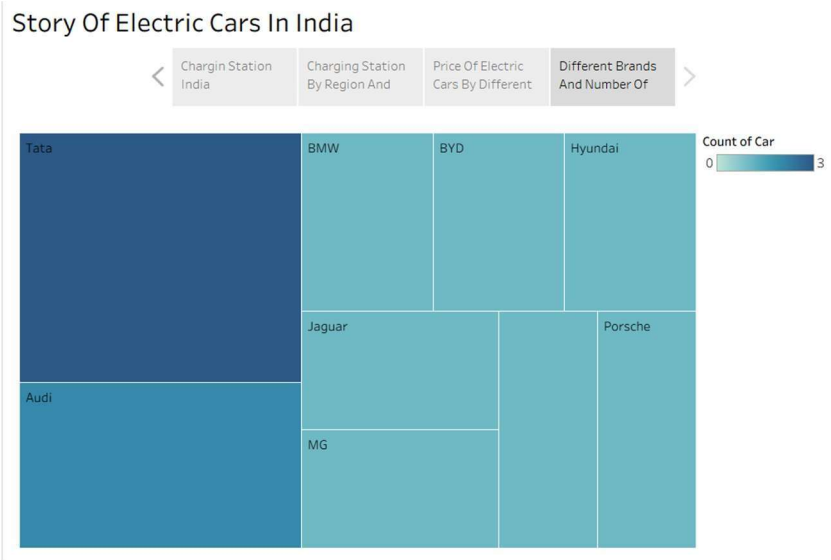

5. Summary

The preprocessing and visualization steps helped transform raw EV datasets into meaningful insights using Tableau dashboards and stories. The analysis enables users to understand EV charging patterns, compare vehicle performance, and make informed decisions.

Tableau Public Link:-
DashBoard:-
https://public.tableau.com/views/Book3_17729455385230/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link